MCDB-Net: Multiview Collaborative Dual-Branch Unmixing Network for Hyperspectral Images

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***Abstract*—Hyperspectral unmixing (HU) based on autoencoders (AEs) is a crucial research task in remote sensing imagery. While existing deep HU networks primarily focus on spatial features, the inherently rich and continuous spectral bands in hyperspectral images (HSIs) harbor significant underutilized information. The intrinsic characteristics of these continuous bands can naturally enable more nuanced and effective modeling of mixed pixels. In this article, we propose the multiview collaborative dual-branch network (MCDB-Net), designed to fully learn and exploit the complex spectral features in high-dimensional hyperspectral data, thereby enhancing the representation capabilities of these features. MCDB-Net constructs a novel multiview spectral block to strengthen the correlation between multiple views of pixel spectra. The full-view spectral feature extraction module highlights important spectral features, while the local multiview spectral feature extraction module provides a detailed understanding of the interactions between multiview spectral information. The multiview abundance collaboration module collaboratively learns spectral feature information from different perspectives and dynamically adjusts the weights of abundance estimations, leading to better integration of abundance features across various views. Extensive experiments on different datasets demonstrate that MCDB-Net achieves higher continuity and robustness in unmixing results, showcasing its powerful capability in spectral feature extraction.**

***Index Terms*—Attention mechanism, autoencoder (AE), hyperspectral image (HSI), multiview learning, spectral unmixing.**

I. Introduction

THE spectral resolution of hyperspectral image (HSI) has significantly improved with technological advancements, increasing the number of spectral bands to hundreds or thousands, thereby revealing intricate spectral details [1], [2]. However, the restricted spatial resolution often results in mixed pixels composed of various ground materials. Hyperspectral unmixing (HU) addresses this challenge by extracting pure spectral signatures and estimating the abundance of different materials, improving material classification and identification while supporting subsequent hyperspectral tasks [3]. This work focuses on linear spectral unmixing (LMM), where mixed pixel spectra are linearly combined to infer material abundances, facilitating material classification and quantitative analysis [1]. Under the constraints imposed by physical conditions, abundances must comply with two crucial constraints: the first is the abundance nonnegativity constraint (ANC); the second is the abundance sum-to-one constraint (ASC) [4].

In this work, we mainly focus on the issue of LMM. This includes traditional linear unmixing methods based on geometry, statistics, and sparse regression, as well as HU methods based on deep learning that have been widely used in recent years. There are mainly two types of HU methods based on geometry: methods based on the pure pixel assumption and methods based on minimum volume (MV). For the unmixing algorithms based on pure pixels, representative algorithms include the vertex component analysis (VCA) [5], the simplex growing algorithm (SGA) [6], and N-FINDR [7]. The fully constrained least squares (FCLSs) algorithm [4] relies on the extracted endmember information to further accurately calculate and obtain the abundance information. However, in practical situations, there are few cases that satisfy the pure pixel assumption, which greatly increases the complexity and challenge of HU. To handle highly mixed HSI, a more efficient MV simplex analysis (MVSA) [8] has been presented. When the spectral signals of HSI are extremely mixed, statistical-based methods become effective choices. Prominent examples of unmixing methods are those based on the Bayesian framework [9], [10], [11] and nonnegative matrix factorization (NMF) [12], [13], [14]. The first approach uses statistical assumptions and prior knowledge to enhance the unmixing performance [10]. In contrast, the second one breaks down the HSI into two nonnegative matrices [15]. This enables the concurrent estimation of endmembers and abundances. To enhance the unmixing precision, researchers propose extended NMFs with regularization restriction [16], [17], [18], [19], [20], [21].

To ensure that all spectral signals have clear physical interpretations and maintain sparsity, a large number of unmixing algorithms incorporating spectral libraries based on sparse regression have emerged. The sparse unmixing algorithm via variable splitting and augmented Lagrangian

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(SUnSAL) [22] imposes l_1-norm sparse constraint on the abundance matrix, which helps to enhance the sparsity of the abundance vectors. The SUnSAL-TV [23] takes into account the relationship between the abundance vectors corresponding to each pixel and its surrounding pixels during the unmixing process by introducing the total variation (TV). The collaborative SUnSAL (CLSUnSAL) [24] adopts a collaborative sparse regression framework and improves the unmixing results by solving a problem of joint sparse regression. In addition, in [25], iterative weighted sparse unmixing methods are innovatively proposed, which optimizes the unmixing results through ingenious design of the iterative process. Meanwhile, in order to explore the abundant spatial and spectral features contained in HSI more comprehensively and deeply, a series of weighted sparse unmixing algorithms have been continuously developed [26], [27], [28], [29], [30]. These algorithms are committed to improving the utilization efficiency of spatial–spectral features, optimizing the unmixing models from different perspectives and aspects, and finally achieving the goal of enhancing the unmixing results.

Over the past few years, deep learning has been widely applied in HU, driving the development of HU. In the autoencoder (AE) network structure, the encoder reduces the dimension of the input data to generate abundance fractions. The weights of the decoder are regarded as endmembers, and meanwhile, pixels are recreated based on the obtained abundances. Typical examples constructed based on this framework include uDAS [31], DAEN [32], MTAEU [33], EndNet [34], TANet [35], SSRDMF [36], DASS-Net [37], A2SAN [38], SSABN [39], RTViT-Net [40], MAT-Net [41], and SSLT-Net [42]. uDAS adopts denoising constraints to avoid introducing additional denoising errors and uses $l_{2,1}$ constraints to reduce the redundancy of the encoder, thus lowering the reconstruction error. DAEN employs a stacked AE (SAE) to learn spectral features, generating a better initialization for the unmixing process. It uses a variational AE (VAE) for blind signal separation to obtain both endmembers and abundances simultaneously. MTAEU uses multitask learning to train multiple AEs in parallel to simultaneously decompose the neighborhoods of pixels, thereby alleviating the impact of endmember translation variability. EndNet effectively applies sparsity and nonlinearity by correcting the ReLU activation function and the normalization layer, and proposes a new loss function to enhance sparsity. TANet adopts two AEs that share the same weights to extract physically meaningful endmembers. SSRDMF reduces the impact of Gaussian noise and sparse noise by explicitly modeling sparse noise and introducing self-supervised constraints to explore spectral information. DASS-Net explores spatial information by leveraging neighborhood attention and fuses spatial and spectral features to improve the accuracy of endmember extraction and abundance estimation. Tao et al. [38] propose the A2SN and A2SAN networks. Through innovative unmixing strategies, embedding attention mechanisms, and introducing an abundance guidance mechanism, these networks can directly extract endmembers from the network, estimate abundances, and reconstruct images, enhancing the robustness against noise. Yang et al. [39] construct the spatial–spectral attention bilateral network. By utilizing Swin transformer blocks and a simplified spectral channel attention mechanism to extract and fuse spatial and spectral information, respectively, the performance of HU is improved. Shu et al. [40] develop the RTViT-Net specifically for HU, integrating an innovative token generation mechanism into its design. To address the limitations of single-scale feature extraction in HU, Wang et al. [41] propose the MAT-Net, which achieves spectral and multiscale spatial feature fusion through a dual-stream multibranch encoder and a customized transformer encoder. Deng et al. [42] introduce SSLT-Net, a hybrid model that combines a convolutional neural network (CNN) and a linear transformer, to realize the optimized fusion of local, global, spatial, and spectral features for HU tasks.

In the field of HU, CNNs have emerged as a high-performance network architecture due to their remarkable local perception capability and superior ability to extract spatial–spectral features [43], while transformers have also been widely applied by virtue of their advantages in global feature modeling. Palsson et al. [44] directly explore the spatial correlations of HSI through CNN and propose a novel spatial–spectral linear mixing model, which simplifies the application of spatial regularization. Gao et al. [45] propose a cycle-consistent unmixing network. By relaxing the original pixel-level reconstruction assumption and imposing cycle-consistency constraints, they transfer high-level semantic information into the AE-based unmixing network. Fang et al. [46] propose a Bayesian fully convolutional unmixing network, which explores the spatial correlations and noise heterogeneity in HSI. Rasti et al. [47] propose a convolutional unmixing network based on a high-depth minimum simplex volume. It utilizes geometric information by adding a minimum simplex volume penalty term to the loss function of endmember extraction and incorporates spatial information using convolutional filters. Qi et al. [48] propose a new dual-stream network for spatial–spectral collaborative unmixing. This network consists of a spatial AE network based on superpixel segmentation and a spectral AE network based on a convolutional network. The two AE networks are trained effectively in a collaborative manner. Meanwhile, relevant literature [49], [50] leverages CNN to efficiently extract profound and valuable feature information from highly complex hyperspectral data, capture the subtle details within high-dimensional data, and significantly improve the accuracy of abundance estimation.

Among the above approaches, the majority of them [23], [26], [27], [44], [45], [46], [47], and [48] primarily explore the spatial information in hyperspectral data. As indicated in [23], [44], and [48], the spatial feature is of great significance in the unmixing process. At the same time, spectral information also plays a crucial role. The spectral resolution of HSIs is gradually increasing, and the number of spectral bands they contain is constantly growing, thus presenting more complex and detailed spectral information. Therefore, fully excavating the information in spectral data and enhancing the ability to express spectral features have become key tasks in the research of HU.

In order to effectively utilize the spectral information of HSI, various spectral partitioning methods have been proposed to enhance the spectral feature representation ability.

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In contrast to conventional band selection approaches, spectral partitioning does not simply select a part of the bands. Instead, it utilizes the whole set of spectral bands to provide multiview data for subsequent image processing by generating multiple views of the HSI. Spectral partitioning divides the spectral bands into multiple highly correlated subsets [51], and regenerates subsamples of the original spectral bands from the obtained spectral clusters. A multitude of spectral partitioning methods exist in the realm of HSI classification. Notable examples include the methods described in [52], [53], [54], and [55]. In [52], a spectrometer-driven (SD) classification strategy for HSI is proposed. It is based on a full understanding of the working principle of spectrometers and the physical characteristics of imaging spectrometers to improve the classification capabilities. In [54], adaptive affinity propagation (AAP) [56] is adopted to automatically select the number of clusters and centering parameters through unsupervised clustering, and reassign the partitions into new band groups. Similarly, there are applications in the field of unmixing. RMSU [57] is a spatial–spectral multiview data generation approach that performs multiview dictionary learning and estimates abundances based on a joint dictionary. In addition, Qi et al. [58] propose the spectral–spatial-weighted multiview collaborative sparse unmixing method to make full use of spectral and spatial information and improve the unmixing performance. This marks the first proposal of the multiview unmixing concept in the field of HU. Experimental validation has revealed that abundance estimates from different views exhibit significant discrepancies, which can be effectively harnessed to enhance the accuracy of abundance estimation. Based on this finding, Qi et al. [59] further introduce the significant discovery of multiview strategies into the AE-based unmixing framework and propose a multiview spatial–spectral two-stream network. By designing two AEs and leveraging recurrent neural networks (RNNs), they achieve the collaborative utilization of multiview spectral–spatial information, enabling the unmixing process to more meticulously explore the potential value of multidimensional spectral feature information.

In summary, most of the existing HU methods extract spectral features from a single perspective of mixed pixels, which severely restricts the effective capture of deep spectral features. As a result, when faced with complex land object scenes, it is difficult to accurately analyze the rich spectral feature information contained in hyperspectral data, especially in the obvious deficiency of modeling complex spectral interactions. Following the development of multiview unmixing, this study focuses on the in-depth mining of spectral features in hyperspectral data and the efficient integration of multiview information, and is committed to improving the performance of multiview unmixing. In the unmixing model based on the AE, a novel multiview spectral construction method is innovatively introduced to fully unleash the technical potential of multiview unmixing. This method divides the spectral bands from multiple views and skillfully combines them into spectral blocks, enabling the constructed model to comprehensively and deeply utilize complex spectral interactions, thus providing a brand-new idea for HU. Based on this, this article proposes an HU model called multiview collaborative dual-branch network (MCDB-Net) based on multiview abundance collaboration. This model is expected to significantly improve the accuracy and robustness of HU. The main contributions of this study are as follows.

1) To break through the limitation that traditional HU methods over-rely on a single view and fail to fully excavate spectral features, a novel multiview unmixing learning framework based on single-pixel expansion is proposed. Abandoning the analytical mode of isolated single-pixel spectra, this framework innovatively transforms individual pixel spectra into multidimensional spectral combinations by constructing multiview spectral blocks, realizing a leap from single-point feature extraction to multidimensional feature collaborative capture. It provides a new idea for the thorough mining of spectral information in complex mixed ground object scenarios, significantly enhances the model's adaptability to complex mixed spectra, and lays a solid foundation for accurate unmixing.
2) To efficiently integrate the global and local multiview spectral information, a dual-branch multiview network structure is adopted. The global branch comprehensively captures the overall spectral features of hyperspectral data through a full-view attention mechanism, suppressing unimportant spectral bands and highlighting key spectral features. The local branch generates multiple subviews based on advanced spectral partitioning techniques and constructs multiview spectral blocks with specific spatial and channel dimensions. By using deep convolutional operations, it accurately captures the local response patterns of the same spectral band at different view positions, exploring the subtle changes and unique features between bands. Meanwhile, the multiview attention mechanism is introduced to adaptively focus on the subtle differences and complex relationships among different local views. The two branches work in coordination to achieve a comprehensive and detailed understanding and analysis of hyperspectral data from the overall to the local level, significantly improving the unmixing accuracy.
3) A general multiview abundance collaboration module is designed for the processing of abundance information under local and global views. Since the spectral features of different ground objects in HSI vary across different views, this module dynamically weights the multiview abundance scores according to these differences. It effectively reduces the discontinuity of the unmixing results at the boundaries of ground objects, making the unmixing results more consistent with the actual distribution characteristics of ground objects.
4) In the field of HU, the correlation mechanism of multiview spectral features is established for the first time, realizing the innovative interaction between global and local spectral information in the unmixing network, which enhances the application potential of multiview unmixing in this field. A large number of experiments

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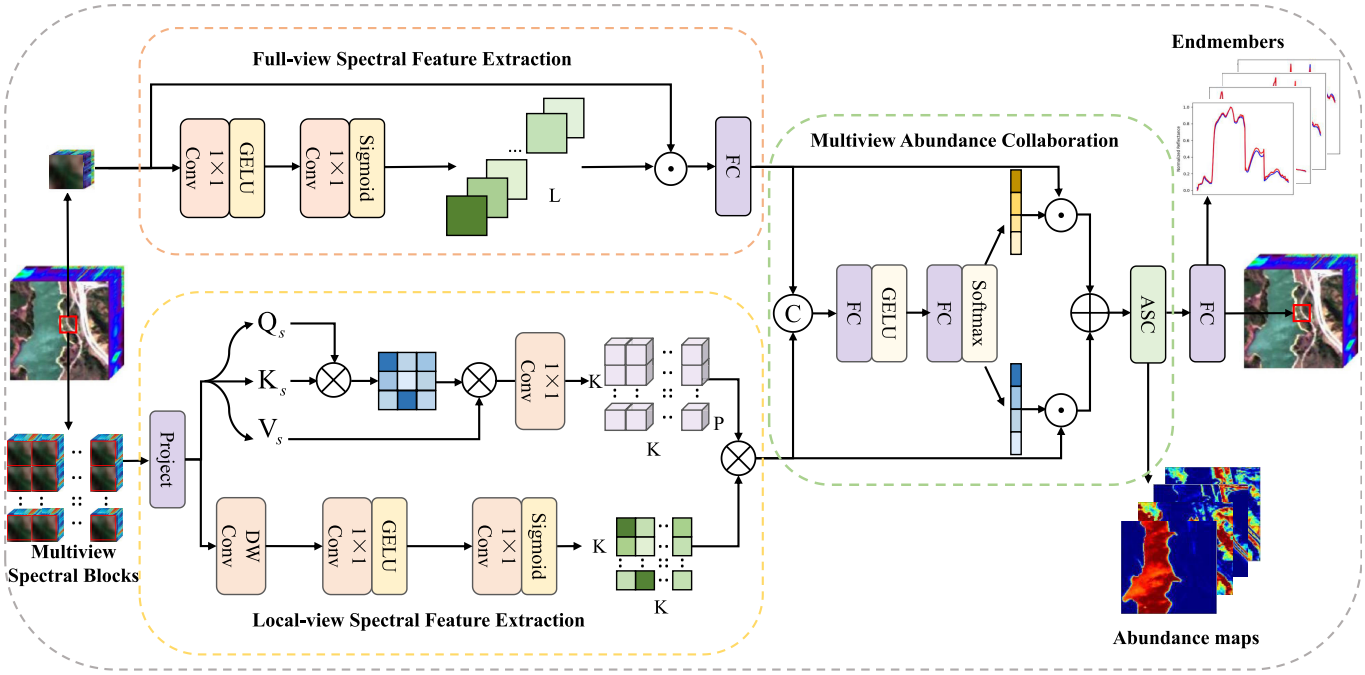


Fig. 1. Schematic of the put forward MCDB-Net architecture.

have demonstrated the superiority of this network on different datasets.

II. Proposed Framework

In order to further mine spectral feature information, explore information interaction under multiple views, integrate global and local feature information, and achieve full utilization of spectral features at the single-pixel level, we propose a novel multiview collaborative dual-branch unmixing network, as shown in Fig. 1. The network as a whole adopts an AE architecture. MCDB-Net, which includes a full-view spectral feature extraction module, a local multiview spectral feature extraction module, and a multiview abundance collaboration module, will be described in detail next.

A. Multiview Data Construction

In the field of HU, accurately dividing spectral bands and efficiently utilizing their information are of utmost importance. Typical spectral partitioning techniques include the AAP [56] algorithm based on statistical principles and the SD [52] strategy based on physics-driven mechanisms. They can be applied to different hyperspectral data types. SD can be used for typical imaging spectrometers, such as AVIRIS [60] and DAIS [61], while AAP can be used as a supplement for other types of data. The AAP algorithm is derived from the AP algorithm and aims to address the inconsistency problem in spectral band search while fully considering the correlations among spectral bands. According to the AAP algorithm, after spectral band division, the bands within the clustered clusters exhibit strong correlations. Let the number of these clusters be t. On the other hand, the spectral division technique based on SD is an instrument-driven spectral division strategy [52] based on a thorough understanding of the working principle of spectrometers and the physical characteristics of imaging spectrometers. Let the number of imaging spectrometers in the hyperspectral imaging system be w, which has the same meaning as the number of clusters divided in the AAP algorithm.

By clustering with the AAP algorithm, t band clusters can be obtained, or by using the SD strategy, w spectral groups can be obtained. Subsequently, an equal-interval band sampling method [52] is adopted to reassign highly correlated spectral bands to q spectral partitions. These q spectral divisions correspond to q perspectives, forming a multiview dataset $\{\mathbf{x}_c^m\}_{m=1}^q$. Here, for a single pixel $\mathbf{x}_c \in \mathbb{R}^L$ in a HSI (L represents the total number of bands of the current pixel), $\mathbf{x}_c^m \in \mathbb{R}^l$ represents the data of the mth perspective under the cth pixel, and each perspective contains l bands.

Suppose $q = K \times K$, that is, the number of perspectives can just be arranged in a matrix form of $K \times K$ (K also represents the window size of the multiview spectral block). We arrange these $K \times K$ perspective data in a 2-D matrix form. Each element in the matrix corresponds to a perspective, and the l bands contained in each perspective are extended along the third dimension. Let the perspective dataset be $\{\mathbf{x}_c^m\}_{m=1}^{K\times K}$. Through the mapping relationship $m = (i-1) \times K + j$, where $i = 1, \ldots, K$ and $j = 1, \ldots, K$, a 3-D multiview spectral block $\mathcal{X} \in \mathbb{R}^{K\times K\times l}$ is constructed, as shown in Fig. 2.

After such spectral partitioning and construction of the multiview spectral block, the spectral band curves of each perspective basically match the spectral curve of the original HSI, and the information between perspectives is complementary, significantly improving the utilization rate of spectral

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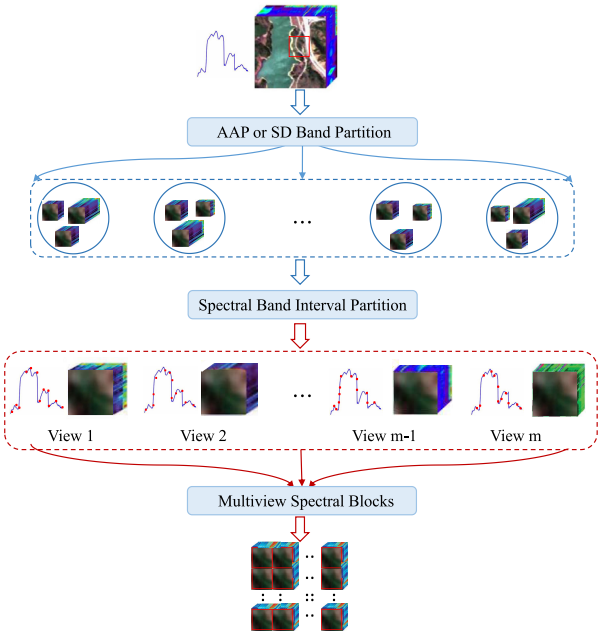


Fig. 2. Construction process of multiview spectral blocks.

information. Through the construction of this multiview spectral block, the orderly integration of multiview spectral information is achieved, providing a more structured and effective data form for subsequent HU processing, and promising to further improve the accuracy and effect of unmixing.

In the dual-branch network structure we constructed, the lower branch takes the above-constructed multiview spectral block as input. The multiview spectral block helps it to effectively extract the local features of mixed pixels through specific spectral division and construction methods. The upper branch directly inputs full-band pixels. Essentially, full-band pixels can be regarded as a special case of the multiview spectral block. In the construction logic of the multiview spectral block, when the number of perspectives is 1×1, that is, $K = 1$, the multiview spectral block at this time is equivalent to the input of full-band pixels. This dual-branch structure, on the one hand, uses the multiview spectral block to excavate local detailed information, and on the other hand, relies on full-band pixels to retain the overall spectral characteristics. The two complement each other, providing strong support for more comprehensive and efficient processing of HSI data, and promising to further optimize the performance of HU.

B. Full-View Spectral Feature Extraction Module

The full-view branch adopts a simple network structure in the encoder design [31], [62]. Due to the inherent limitations of HSI data, such as limited sample quantity and relatively single class distribution, as stated in [63], blindly deepening the coding layer of the unmixing network not only makes it difficult to improve the accuracy of endmember extraction but also triggers the risk of overfitting due to a significant increase in the number of parameters. At the same time, to reduce the complexity of the network, for the full-band information of HSI, a per-pixel input $\mathbf{x}_c$ is adopted. As shown in Fig. 1, a full-view attention module can introduce additional nonlinear spectral information to complement the feedforward neural network (FFN), reduce spectral band redundancy, enhance spectral features, and achieve enhancement of spectral band information. Given a fixed single-pixel input $\mathbf{x}_c \in \mathbb{R}^L$, it can be expressed as

$$\mathbf{G}_{full} = f\left(\mathbf{W}_2\sigma\left(\mathbf{W}_1\mathbf{x}_c\right)\right) \tag{1}$$

where $f(\cdot)$ is the Sigmoid function, $\sigma(\cdot)$ denotes the GELU function, and $W(\cdot)$ represents the weights of pointwise convolution after spectral band dimension reduction. $\mathbf{W}_1$ represents the weight values after reducing the proportion r_1, while $\mathbf{W}_2$ represents the weight values after increasing the proportion r_1.

Subsequently, the obtained full-view attention map $\mathbf{G}_{full}$ is applied to the initial input $\mathbf{x}_c$, thereby achieving enhancement of the spectral bands, a process that can be described as

$$\hat{\mathbf{x}}_c = \mathbf{x}_c \odot \mathbf{G}_{full}. \tag{2}$$

After the spectral bands are enhanced by the full-view attention, they pass through a fully connected layer to achieve encoding of the global spectral bands, obtaining coarse-grained abundance fractions under the full-view perspective, which can be denoted as

$$\mathbf{H}_{full} = \mathrm{ReLU}\left(\mathrm{Dropout}\left(\mathrm{BN}\left(\mathbf{W}_3\hat{\mathbf{x}}_c\right)\right)\right) \tag{3}$$

the ReLU activation function is used to enforce the nonnegativity constraint on abundances. In addition, BN denotes the batch normalization layer, and Dropout is used to prevent overfitting, where $\mathbf{W}_3$ is the weight coefficient of a fully connected layer without bias. $\mathbf{H}_{full}$ represents the coarse-grained abundance obtained from the HSI through the full-view branch, with a dimension of $1 \times P$ (where P denotes the number of endmembers), and it serves as one of the inputs for multiview abundance fusion.

C. Local Multiview Spectral Feature Extraction Module

Currently, HU networks typically use single-view input and do not consider the relationships between different perspectives. This work mainly adopts the selection of equal-interval bands to generate a spectral set of multiview data. For the divided multiview datasets, a multiview spectral block is constructed, and then input into the designed dual-stream network. The main idea is to capture the connections between different perspectives through the designed multiview attention mechanism, use pointwise convolution to encode the spectral bands, and generate coarse-grained abundance fractions under the current perspective. Second, by combining deep separable convolution with pointwise convolution in series, local spectral context information is combined to extract local spectral features and guide the generation of multiview abundance weights.

As shown in Fig. 1, multiview attention is utilized for encoding to achieve the interaction among multiview datasets. Each view contains information from other views, thus promoting the continuity of abundance under different views. For the input multiview spectral block $\mathcal{X}$, linear projection encoding

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is used to generate $\mathbf{Q_s}, \mathbf{K_s}, \mathbf{V_s}$ matrices, all of which have dimensions in $\mathbb{R}^{K\times K\times l}$, which can be formulated as

$$\mathbf{Q_s} = \mathcal{X}\mathbf{W}_Q, \quad \mathbf{K_s} = \mathcal{X}\mathbf{W}_K, \quad \mathbf{V_s} = \mathcal{X}\mathbf{W}_V \tag{4}$$

where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{l\times l}$ are obtained through unbiased linear mapping. After reshaping, the matrices are represented as $\mathbf{Q}_s, \mathbf{K}_s, \mathbf{V}_s \in \mathbb{R}^{K^2\times l}$, and the final output after reshaping is $\mathbf{Y}_s \in \mathbb{R}^{K\times K\times l}$, as follows:

$$\mathbf{Y}_s = \text{Reshape}\left(\text{Softmax}\left(\mathbf{Q}_s\left(\mathbf{K}_s\right)^T/\beta\right)\cdot \mathbf{V}_s\right). \tag{5}$$

After multiview attention, the relationships among multiple views are established, and then pointwise convolution is utilized to obtain the abundance fractions under multiple views, which can be represented as

$$\mathbf{h}_{\text{loc}} = \sigma\left(\mathbf{W}_4\mathbf{Y}_s\right) \tag{6}$$

the dimension of $\mathbf{h}_{loc}$ is $K \times K \times P$ (where P denotes the number of endmembers), and $\mathbf{W}_4$ is the weight coefficient of an unbiased fully connected layer.

For the input multiview spectral block, local spectral feature extraction is performed to generate dynamic attention weights for local multiview abundance fusion, which is given as

$$\mathbf{d}_w = f\left(\mathbf{W}_6\sigma\left(\mathbf{W}_5\text{DW-Conv}\left(\mathbf{V_s}\right)\right)\right) \tag{7}$$

where $\mathbf{W}_5$ and $\mathbf{W}_6$ are the weight values of two pointwise convolutions. Before the pointwise convolutions, a channel-wise convolution (DW-Conv) is applied to extract local features of the multiview spectral blocks. After the first pointwise convolution, the spectral dimension is reduced to l/r_2, and after the second layer, it is reduced to 1. Finally, after the Sigmoid activation function, a multiview weight matrix of $K \times K \times 1$ is obtained. $\mathbf{d}_w$ is the dynamic weight distribution of the generated multiview spectral blocks under different views, which is used to further fuse the coarse-grained abundances from local multiview.

For the generated multiview abundance $\mathbf{h}_{loc}$, dynamic weight fusion is performed, which is defined as

$$\mathbf{H}_{\text{loc}} = \mathbf{d}_w \cdot \mathbf{h}_{\text{loc}}. \tag{8}$$

$\mathbf{H}_{loc}$ represents the local multiview abundance after dynamic weighting, with a reshape dimension of $1 \times P$. It serves as the final abundance of the current pixel and is used for subsequent local–global abundance fusion.

D. Multiview Abundance Collaboration Module

We design a multiview abundance collaboration module to achieve dynamic fusion of global and local multiview abundance features, thereby obtaining more accurate abundances, eliminating abundance edge discontinuities, and making full use of spectral information under different perspectives.

The full-view abundance $\mathbf{H}_{\text{full}}$ and the local-view abundance $\mathbf{H}_{\text{loc}}$ are concatenated, with both having the same dimension, which can be specifically expressed as

$$\mathbf{H} = \mathbf{H}_{full} \oplus \mathbf{H}_{\text{loc}}. \tag{9}$$

After concatenation, the perspective abundance is input into the dynamic abundance collaboration module, which consists of two steps: abundance collaboration and weight allocation. The abundance collaboration module achieves a high degree of fusion of local and global abundances through a fully connected layer, and the collaboration step can be expressed as

$$y_{\text{fuse}} = \sigma\left(F_{fc_1}\left(\mathbf{H}\right)\right) \tag{10}$$

where F_{fc_1} is a fully connected network layer, $\sigma(')$ is the GELU activation function, and the fused abundance $y_{fuse} \in \mathbb{R}^{1\times 1\times P'}$, and P' is set to $P/2$.

Subsequently, the collaborated data is sequentially processed through a fully connected layer and a Softmax activation function to generate weight coefficients corresponding to different perspectives. The core of this step lies in dynamically allocating weights to automatically assign higher weights to the superior parts of abundance features in either the local or global perspective. Specifically, when a certain perspective demonstrates more accurate abundance performance in specific regions, its weight will be correspondingly increased. Through this mechanism, adaptive collaboration of multiview abundances is achieved, fusing the optimal abundance results from different branches to attain the optimal outcome, which can be specifically expressed as

$$\begin{aligned}\alpha_f &= \frac{e^{F_{\alpha_f}(y_{\text{fuse}})}}{e^{F_{\alpha_f}(y_{\text{fuse}})} + e^{F_{\alpha_{\text{loc}}}(y_{\text{fuse}})}}\\ \alpha_{\text{loc}} &= \frac{e^{F_{\alpha_{\text{loc}}}(y_{\text{fuse}})}}{e^{F_{\alpha_f}(y_{\text{fuse}})} + e^{F_{\alpha_{\text{loc}}}(y_{\text{fuse}})}}.\end{aligned} \tag{11}$$

The abundance under the full view and the local view are dynamically weighted and summed to obtain the final abundance of the current pixel $\hat{\mathbf{y}}_s$

$$\hat{\mathbf{y}}_s = \alpha_f \odot \mathbf{H}_{full} + \alpha_{\text{loc}} \odot \mathbf{H}_{\text{loc}}. \tag{12}$$

Meanwhile, to satisfy the constraints of abundance and uniqueness, the l_1-norm is adopted in this implementation. Therefore, the final pixel abundance can be presented as follows:

$$\mathbf{s}_c = \frac{\hat{\mathbf{y}}_s}{\|\hat{\mathbf{y}}_s\|_1 + \varepsilon} \tag{13}$$

where ε is a very small positive number, mainly to prevent the denominator from being meaningless.

Finally, the abundance obtained is passed through the decoder to reconstruct the current pixel $\hat{\mathbf{x}}_c$. Based on the LMM formula, the following can be obtained:

$$\hat{\mathbf{x}}_c = \mathbf{W}^{(d)}\mathbf{s}_c \tag{14}$$

where $\mathbf{W}^{(d)}$ represents the weight coefficient of the fully connected layer of the decoder, and its corresponding values are the endmembers that need to be extracted in the HU task.

E. Initialization and Loss Function

In the research of spectral unmixing, the advantages of the spectral angle distance (SAD) have been verified by numerous papers [34], [44], [63]. Its scale invariance enables it to robustly measure the difference between the original and reconstructed spectra in the unmixing task. Moreover, the SAD function has been successfully applied in the unmixing

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research based on the AE. Therefore, we adopt SAD as the basic loss function. At the same time, considering the key feature of the sparsity of the abundance fractions, we introduce the $l_{1/2}$ norm in the design of the loss function. Compared with the traditional l_1 norm, the $l_{1/2}$ norm has a stronger shrinking effect on nonzero elements, which can induce sparser solutions and obtain sparser abundance vectors. Therefore, the overall loss of MCDB-Net can be represented as follows:

$$L_{\mathrm{MCDB}} = \lambda_1 \mathbf{S}AD\langle \mathbf{x}_c, \hat{\mathbf{x}}_c \rangle + \lambda_2 |\mathbf{s}_c|_{\frac{1}{2}} \tag{15}$$

where $\mathbf{x}_c$ and $\hat{\mathbf{x}}_c$ represent the original pixel and the reconstructed pixel, separately, $\mathbf{s}_c$ represents the abundance vector of the original pixel obtained after fusing the full view and local multiview abundance through the multiview abundance collaboration module, while λ_1 and λ_2 are regularization parameters.

III. Experiments

We employ seven comparative algorithms to validate the effectiveness of the multiview abundance collaboration network. Traditional methods include VCA [5] and $L_{1/2}$-NMF [12]. Five deep-learning-based methods are EndNet [34], TANet [35], CNNAEU [44], MAT-Net [41], and MSSR-Net [59]. It should be particularly noted that, in this comparative experiment, the multiview branch of MSSS-Net we selected is MSSR-Net. By comparing with the aforementioned algorithms, the multiview spectral block construction method proposed in this article can more fully explore spectral features, providing more informative input for the unmixing task.

A. Performance Metrics and Data Description

To thoroughly evaluate the performance of the proposed network, we utilize the SAD, a metric designed to determine the endmembers, and the root mean square error (RMSE), a measure employed to assess the accuracy of abundance estimates. These metrics are the most used in this context and are formulated as follows:

$$SAD = \arccos \frac{< \mathbf{x}_i, \hat{\mathbf{x}}_i >}{\|\mathbf{x}_i\|_2 \, \|\hat{\mathbf{x}}_i\|_2} \tag{16}$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (\mathbf{s}_i - \hat{\mathbf{s}}_i)^2}. \tag{17}$$

In both metrics, a lower value indicates a better outcome. The SAD and RMSE experimental data were compiled from twenty iterations, with the mean values recorded.

We utilize three real-world datasets (Jasper Ridge, Samson, and Cuprite) along with a synthetic dataset to comprehensively evaluate the effectiveness of the proposed method, as illustrated in Fig. 3. The datasets underwent preprocessing steps consistent with those described in the literature [12], [15], [19], and [64].

1) Synthetic Dataset: The synthetic dataset adopted in this study is sourced from the work of Tang et al. [65], which features image patches with a spatial resolution of 64 × 64 pixels, contains no pure pixels, and has been widely adopted as an experimental validation dataset in numerous relevant studies [58], [59], [66], [67]. As illustrated in Fig. 3(a), the

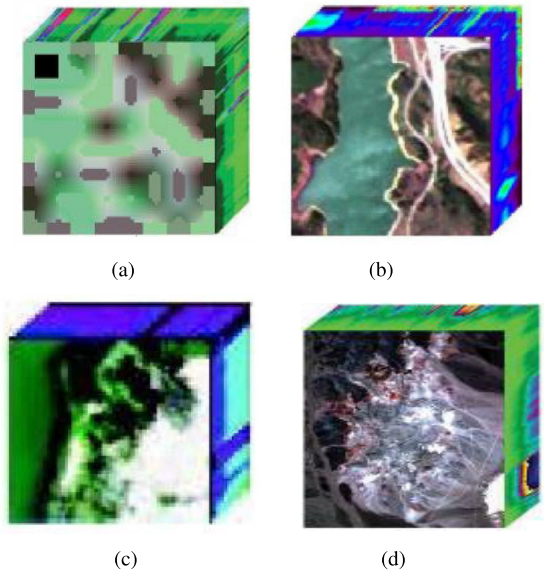


Fig. 3. Schematic of the experimental dataset. (a) Synthetic. (b) Jasper Ridge. (c) Samson. (d) Cuprite.

dataset demonstrates distinct spatial structuring and homogeneous characteristics. The five endmembers incorporated herein are selected from the USGS spectral library [68], with their abundance maps satisfying both the ANC and ASC. In addition, the dataset is contaminated by independent and identically distributed Gaussian noise, which is introduced at varying signal-to-noise ratio (SNR) levels to simulate different degradation scenarios.

2) Jasper Ridge Dataset: The Jasper Ridge dataset was constructed by Stanford University in the United States using the AVIRIS HSI. The original Jasper Ridge data contains a total of 512 × 614 pixels and 224 spectral bands in total. Affected by complex environments such as the atmosphere and water vapor, it fails to truly reflect the conditions of ground objects. Therefore, in this experiment, some spectral bands were removed, and finally, 198 spectral bands were retained. At the same time, 100 × 100 mixed pixel data were selected. There are four types of endmember data in the Jasper Ridge dataset: soil, road, tree, and water.

3) Samson Dataset: The Samson dataset is obtained by the United States Department of Agriculture through the SAMSON sensor. Initially, this dataset contains 952 × 952 mixed pixels, and each pixel is composed of 156 spectral bands. Due to the large number of mixed pixels in this dataset, which leads to high computational costs, mixed pixels within a spatial range of 95 × 95 are selected as the dataset for the current network. The Samson dataset mainly consists of three types of ground objects: soil, water, and tree.

4) Cuprite Dataset: The Cuprite dataset is acquired using the AVIRIS sensor over the Cuprite region in Nevada, USA, with a spatial dimension of 250 × 190 pixels. Following the elimination of noisy and water absorption spectral channels, 188 bands remain for the spectral unmixing analysis. As depicted in Fig. 3(d), the dataset is represented as a 3-D cube, featuring 12 distinct mineral endmembers. These endmembers include Alunite, Andradite, Buddingtonite, Dumortierite,

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TABLE I
HYPERPARAMETERS SETTING

Hyperparameters	Datasets			
	Synthetic	Jasper Ridge	Samson	Cuprite
λ_1	12			
λ_2	$2e-3$			
Batch size	128			
Optimizer	Adam			
Maxiter	200			
Learning rate (encoder)	$1e-3$			
Learning rate (decoder)	$1e-4$	$1e-6$	$1e-4$	$1e-5$

Kaolinite1, Kaolinite2, Muscovite, Montmorillonite, Nontronite, Pyrope, Sphene, And Chalcedony.

B. Experiments on MCDB-Net

In the proposed MCDB-Net, investigating the contribution of different modules to the results is extremely important. We conduct ablation experiments on three modules: the full-view spectral feature extraction module, the local multiview spectral feature extraction module, and the multiview abundance collaboration module. Meanwhile, experiments are performed to analyze relevant parameters within the network, including the dropout probability and the size of the multiview spectral block patch dimension K. To obtain the optimal experimental results, a series of related experiments have been designed.

1) Parameters and Complexity Analysis: In the experiments of this article, based on a large number of experiments and empirical values, the settings related to model training are as follows: The number of model training epochs (Epochs) affects the performance and generalization ability of the network model. Through extensive experiments, it has been verified that setting the maximum number of training epochs to 200 yields the optimal model performance. The learning rates of the encoder and decoder are different. On both the Jasper dataset and the Samson dataset, the learning rate of the encoder is set to $1e-3$. Due to differences in dataset scale and the complexity of ground objects, the decoder weight for the Jasper dataset is $1e-6$, and for the Samson dataset, it is $1e-4$. The regularization parameters λ_1 and λ_2 of the loss function are set the same for both the Jasper Ridge dataset and the Samson dataset, which are 12 and $2e-3$, respectively. The network is implemented under the PyTorch 1.0 deep-learning framework, and the code is written in the Python 3.6 environment. The summarized hyperparameter data are shown in Table I.

The number of parameters is one of the important indicators for evaluating deep-learning models. In order to comprehensively compare MCDB-Net with five deep-learning comparative algorithms, namely, EndNet, CNNAEU, TANet, MAT-Net, and MSSR-Net, a detailed analysis is conducted on the number of their parameters, and the comparison results are summarized in Table II. It is worth noting that MCDB-Net efficiently utilizes a limited number of parameters while maintaining the effectiveness and performance robustness of the unmixing model.

2) Module Contribution Analysis: We conduct ablation studies on a synthetic dataset (with SNR set to 40 dB) and two real-world datasets, and the results are shown in Table III. It can be observed that the network combining global and local multiview spectral features performs best on the Jasper Ridge, Samson real-world datasets, and the synthetic dataset. In addition, the dynamic fusion of abundances through the abundance synergy module significantly reduces the RMSE, demonstrating the importance of multiview partitioning and spectral block construction for spectral feature mining.

TABLE II
COMPARISON OF THE PARAMETER SCALES OF THE UNMIXING MODELS

Networks	EndNet	CNNAEU	TANet	MAT-Net	MSSR-Net	MCDB-Net
#Params	1.0K-1.5K	1.0K-1.5K	15K	17K	9K	12K

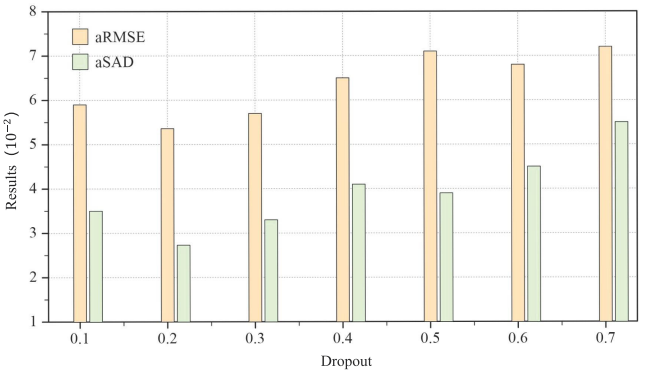


Fig. 4. Quantitative investigation into the degree of dropout.

3) Dropout Strange Analysis: In HU networks, the AE-based network structure plays a leading role, and the magnitude of the hyperparameter α is crucial to the model's performance. In Fig. 4, it is shown that when the batch size is 128 and α ranges from [0.1, 0.7], the MCDB-Net evaluates the endmembers and abundances on the Jasper Ridge dataset. The dropout rate $\alpha = 0.2$ yields the best model performance, effectively balancing regularization and learning capabilities, and optimizing the extraction of endmembers and abundances.

4) Spectral Block Size Analysis: When constructing multiview spectral blocks of size $K \times K$, the selection of parameter K and two different spectral partitioning strategies (AAP and SD) significantly affect the unmixing performance of MCDB-Net. In the parameter setting phase, K is set to [1, 2, 3, 4, 5] to deeply explore the impact of the number of multiview on the unmixing effect of MCDB-Net, as it is directly related to the network's ability to capture and utilize complex spectral information in HSI. As shown in Fig. 5, with the increase of the multiview spectral block K, the abundance evaluation index RMSE shows a trend of increasing rather than decreasing. The network performs best when $K = 2$. Meanwhile, it can also be clearly observed that the spectral band division strategy based on SD outperforms the method based on AAP band division in all cases.

C. Comparison of MCDB-Net Against Other Methods

1) Experiments on Synthetic Dataset: To verify the anti-interference performance and scene adaptation capability of

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TABLE III

ABLATION STUDY INVOLVING SAD AND RMSE ($\times 10^{-2}$) FOR EVALUATING THE CONTRIBUTION DEGREES OF THE DIFFERENT MODULES OF MCDB-NET

Datasets	Full-view	Multiview	Abu Collaboration	aSAD	aRMSE
Jasper Ridge	✓	×	×	3.56 ± 0.1	6.34 ± 0.3
	×	✓	×	3.44 ± 0.2	5.89 ± 0.4
	✓	✓	✓	**2.73 ± 0.1**	**5.36 ± 0.2**
Samson	✓	×	×	3.01 ± 0.3	5.76 ± 0.1
	×	✓	×	2.76 ± 0.1	5.43 ± 0.2
	✓	✓	✓	**2.35 ± 0.1**	**5.13 ± 0.2**
Synthetic	✓	×	×	4.33 ± 0.3	9.26 ± 0.1
	×	✓	×	3.96 ± 0.1	8.89 ± 0.2
	✓	✓	✓	**3.56 ± 0.1**	**8.17 ± 0.2**

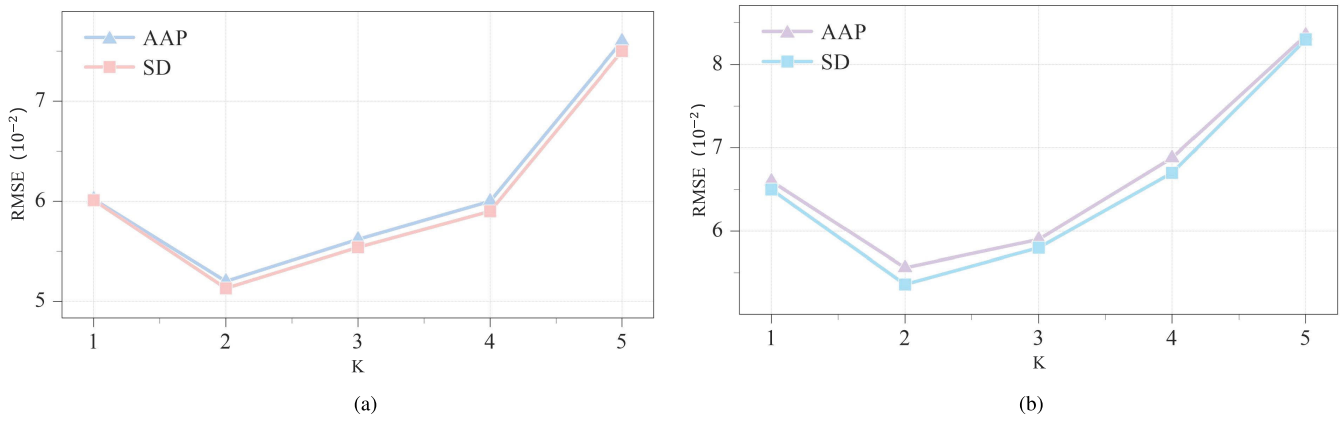


Fig. 5. Quantitative analysis of spectral block sizes via AAP and SD strategies on (a) Samson and (b) Jasper Ridge datasets.

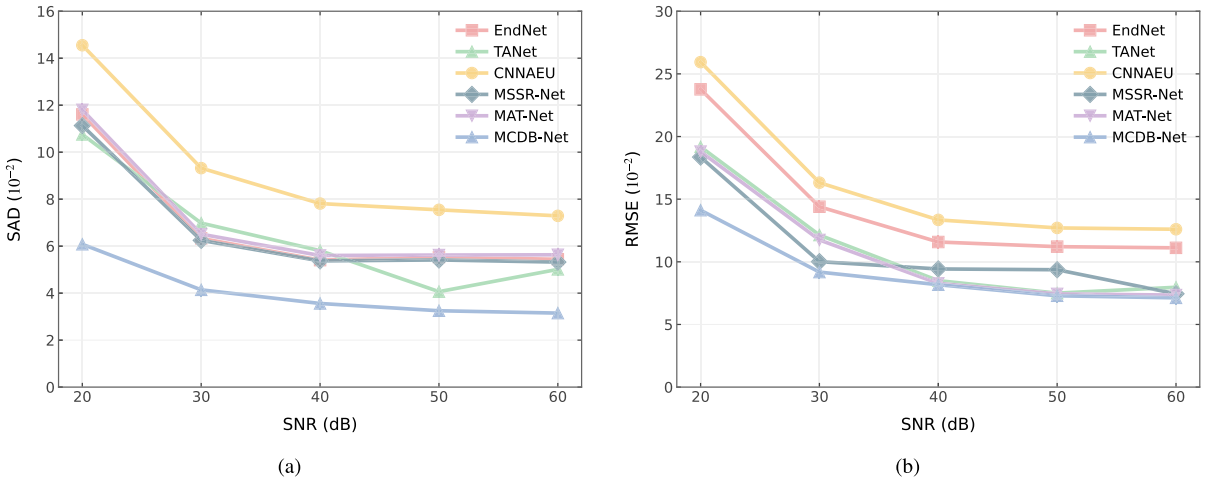


Fig. 6. Performance of various unmixing algorithms across multiple noise intensity levels. (a) Mean SAD. (b) Mean RMSE.

the proposed MCDB-Net in HU tasks, this study conducts multiple sets of controlled experiments on the synthetic dataset. The experiments are configured with five SNR levels: 20, 30, 40, 50, and 60 dB, covering the full range from low to strong noise scenarios to comprehensively evaluate the model's environmental adaptability. The endmember extraction and abundance estimation results of MCDB-Net and five benchmark unmixing methods (EndNet, TANet, CNNAEU, MSSR-Net, and MAT-Net) under different SNR conditions are presented in Fig. 6(a) and (b), respectively. It should be noted that the traditional VCA algorithm and $L_{1/2}$-NMF exhibit a sharp performance decline in low-SNR environments, with their unmixing results deviating significantly from the true values, so they are not included in this comparative analysis.

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TABLE IV

FOR THE JASPER RIDGE DATASET, THE EVALUATION METRIC FOR ENDMEMBERS IS SAD ($\times 10^{-2}$). THE OPTIMAL RESULTS OBTAINED FROM THESE EVALUATIONS ARE MARKED IN BOLD

Endmember(SAD)	VCA	$L_{1/2}$-NMF	EndNet	CNNAEU	TANet	MAT-Net	MSSR-Net	MCDB-Net
Tree	13.95 ± 3.1	15.10 ± 0.3	4.57 ± 0.4	11.94 ± 2.1	**2.83 ± 0.2**	4.89 ± 0.2	4.73 ± 0.1	3.49 ± 0.2
Water	29.09 ± 10.2	4.60 ± 0.0	5.05 ± 0.9	6.92 ± 0.4	3.31 ± 0.3	4.67 ± 0.1	5.60 ± 0.1	**3.12 ± 0.1**
Soil	15.22 ± 2.3	6.16 ± 0.5	5.29 ± 0.3	10.15 ± 0.9	8.48 ± 0.4	3.62 ± 0.2	3.32 ± 0.2	**2.15 ± 0.2**
Road	9.94 ± 1.1	9.81 ± 0.1	3.54 ± 0.2	7.45 ± 0.4	**2.14 ± 0.2**	5.38 ± 0.3	4.15 ± 0.2	2.16 ± 0.1
Average	17.05 ± 3.6	7.19 ± 2.4	4.61 ± 0.5	9.12 ± 0.6	4.19 ± 0.3	4.64 ± 0.1	4.45 ± 0.1	**2.73 ± 0.1**

TABLE V

FOR THE JASPER RIDGE DATASET, THE EVALUATION METRIC FOR ABUNDANCE IS RMSE ($\times 10^{-2}$). THE OPTIMAL RESULTS OBTAINED FROM THESE EVALUATIONS ARE MARKED IN BOLD

Abundance(RMSE)	VCA	$L_{1/2}$-NMF	EndNet	CNNAEU	TANet	MAT-Net	MSSR-Net	MCDB-Net
Tree	16.09 ± 3.1	16.16 ± 0.5	8.85 ± 0.4	13.56 ± 0.3	8.01 ± 0.4	7.63 ± 0.2	6.43 ± 0.4	**2.48 ± 0.4**
Water	6.06 ± 1.8	5.57 ± 0.0	6.88 ± 0.3	9.66 ± 0.3	8.12 ± 0.3	10.12 ± 0.2	7.42 ± 0.4	**4.80 ± 0.2**
Soil	15.00 ± 1.5	17.02 ± 0.4	10.59 ± 0.2	10.61 ± 0.2	6.19 ± 0.4	**3.76 ± 0.4**	7.64 ± 0.3	6.25 ± 0.1
Road	11.09 ± 1.5	6.73 ± 0.2	11.17 ± 0.4	8.64 ± 0.2	**5.45 ± 0.3**	7.21 ± 0.1	9.96 ± 0.3	7.92 ± 0.3
Average	12.05 ± 1.5	11.37 ± 0.2	9.37 ± 0.5	10.62 ± 0.1	7.54 ± 0.3	7.18 ± 0.1	7.86 ± 0.2	**5.36± 0.2**

As observed from the experimental curves, the mean SAD and RMSE values of all tested methods show a downward trend as SNR increases, indicating that the improvement of spectral data quality can effectively drive the optimization of unmixing accuracy. Among all benchmark methods, CNNAEU exhibits the highest SAD and RMSE values in the 20 dB low-SNR scenario, reflecting an obvious limitation in its feature extraction capability for noisy, low-quality HSIs. The proposed MCDB-Net demonstrates superior performance over other comparative models across the entire SNR range, both in endmember extraction and abundance estimation tasks. This advantage stems from MCDB-Net's multiview spectral patch construction mechanism, which enhances the model's ability to mine effective spectral features from noisy data. Meanwhile, its multiview collaborative learning framework efficiently fuses local and global spectral features, significantly boosting the model's generalization and stability under complex noise.

2) Experiments on Jasper Ridge Dataset: Table IV presents the performance comparison results of MCDB-Net with seven other advanced comparison methods on the endmember extraction task of hyperspectral data. The results clearly show that MCDB-Net is superior to the comparison methods in the extraction of Soil and Water endmembers, and in the final endmember evaluation, it improves by 1.46 percentage points compared to the best TANet. Meanwhile, among the seven comparison methods, TANet adopts a dual-stream network structure and shows the most significant extraction effect on Tree and Road endmembers. Through careful analysis, it can be known that the success of TANet is mainly due to the synergistic learning of the dual-stream architecture coupled with the effective utilization of pure and nearly pure pixel information, which greatly enhances the ability to extract physically meaningful endmembers and achieve accurate and interpretable unmixing in hyperspectral data processing. In contrast, although MCDB-Net performs more excellently on the Soil and Water endmembers, it is slightly inferior to TANet in terms of Road and Tree endmembers. This indicates the possible performance differences of different network architectures when dealing with different types of endmember extraction tasks in hyperspectral data, and the performance of HU can be further improved through network architecture innovation.

A detailed analysis of the abundance estimation results of different unmixing models is carried out, as shown in Table V. It is worth noting that although EndNet far outperforms CNNAEU in endmember extraction, the difference between the two in abundance estimation is not significant. This indicates that there is not a strong correlation between endmember extraction and abundance estimation to a certain extent. Excellent endmember extraction does not necessarily guarantee good abundance estimation, which depends on the network model's exploration of spatial–spectral features. Meanwhile, MAT-Net shows significant advantages among the seven comparison methods, especially in the abundance estimation of soil ground objects, slightly higher than the MCDB-Net. It is emphasized that MCDB-Net fully explores the spectral features in HSI, fuses the coarse-grained abundance information from different views, and conducts collaborative learning. It has achieved remarkable results in the abundance estimation of the Tree and Water endmembers, and its overall performance exceeds the seven comparison methods, demonstrating a strong ability in abundance estimation. This further confirms the excellent performance of MCDB-Net in processing hyperspectral data and the unmixing task.

The comparison between the endmember curves presented by MCDB-Net and the true endmember curves on the Jasper Ridge dataset is shown in Fig. 7. In the figure, the blue curves

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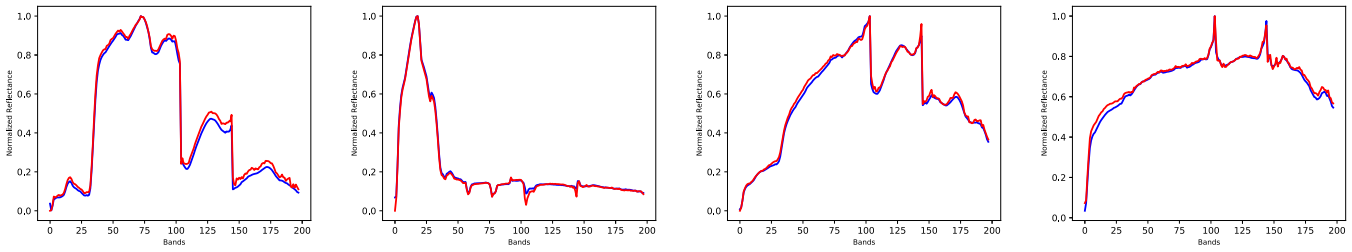


Fig. 7. For Jasper Ridge dataset: MCDB-Net estimated endmembers (blue curves) and GT endmembers (red curves). (a) Tree. (b) Water. (c) Soil. (d) Road.

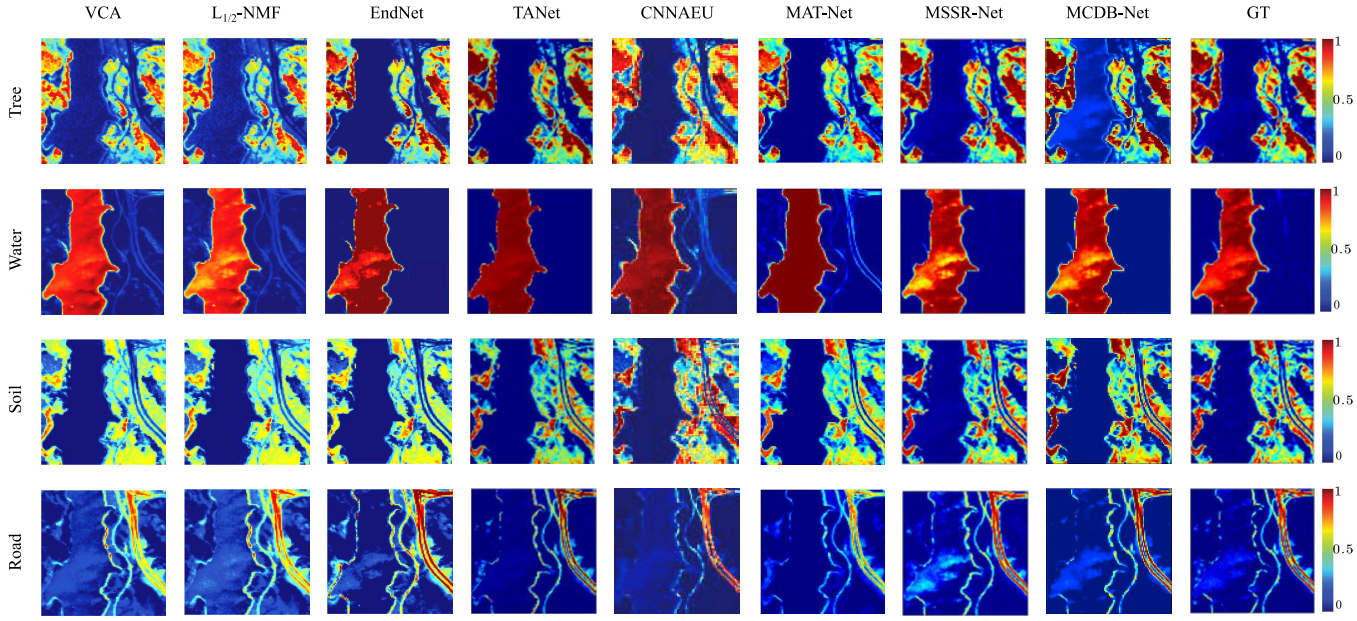


Fig. 8. For Jasper Ridge dataset: abundance maps estimated by different methods and the relevant GTs.

TABLE VI

FOR THE SAMSON DATASET, THE EVALUATION METRIC FOR ENDMEMBERS IS SAD ($\times 10^{-2}$). THE OPTIMAL RESULTS OBTAINED FROM THESE EVALUATIONS ARE MARKED IN BOLD

Endmember(SAD)	VCA	$L_{1/2}$-NMF	EndNet	CNNAEU	TANet	MAT-Net	MSSR-Net	MCDB-Net
Soil	4.21 ± 0.0	6.21 ± 7.3	1.98 ± 0.2	3.73 ± 2.1	1.22 ± 0.2	**1.17 ± 0.1**	2.25 ± 0.1	1.51 ± 0.1
Tree	5.58 ± 0.2	5.23 ± 0.3	5.31 ± 0.3	3.97 ± 0.4	3.31 ± 0.3	3.06 ± 0.2	4.46 ± 0.1	**2.83 ± 0.1**
Water	43.97 ± 31.1	11.97 ± 2.1	3.98 ± 0.2	4.30 ± 0.9	6.22 ± 0.3	3.78 ± 0.2	4.17 ± 0.2	**2.70 ± 0.2**
Average	17.92 ± 1.5	7.80 ± 3.2	3.76 ± 0.2	4.00 ± 0.7	3.58 ± 0.3	2.67 ± 0.2	3.63 ± 0.1	**2.35 ± 0.1**

represent the endmember curves extracted by the method in this article. They almost perfectly fit the curves of Water and Soil, highlighting the excellent endmember extraction effect. Although the curve shapes of the other two types of ground objects also approximate the true endmember curves, there are slight deficiencies in detail. Meanwhile, Fig. 8 shows the visualized abundance estimation results, which highly coincide with the ground truth (GT) in the highlighted parts of Tree and Water ground objects. This further confirms that the proposed MCDB-Net is more accurate in abundance estimation scores. This high degree of coincidence indicates that MCDB-Net has significant advantages in the abundance estimation of complex ground objects and provides a solid foundation for the accuracy of ground object classification and abundance estimation.

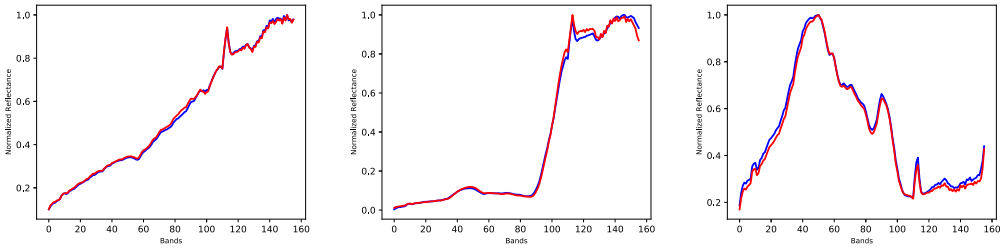
3) Experiments on Samson Dataset: Table VI summarizes the endmember extraction results of all the compared unmixing models and MCDB-Net on the Samson dataset. MAT-Net achieves excellent performance in endmember extraction on this dataset, demonstrating strong model potential, and it performs particularly well in the extraction task of the Soil endmember. In contrast, although MCDB-Net's endmember extraction result for the Soil ground object is slightly lower than that of MAT-Net, its overall endmember extraction effect

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TABLE VII

FOR THE SAMSON DATASET, THE EVALUATION METRIC FOR ABUNDANCE IS RMSE ($\times 10^{-2}$). THE OPTIMAL RESULTS OBTAINED FROM THESE EVALUATIONS ARE MARKED IN BOLD

Abundance(RMSE)	VCA	$L_{1/2}$-NMF	EndNet	CNNAEU	TANet	MAT-Net	MSSR-Net	MCDB-Net
Soil	16.45 ± 3.0	8.58 ± 3.3	8.60 ± 0.0	18.49 ± 0.6	6.96 ± 0.3	**5.72 ± 0.2**	5.84 ± 0.3	5.79 ± 0.3
Tree	11.25 ± 1.0	7.44 ± 3.7	6.92 ± 0.1	16.21 ± 0.4	5.71 ± 0.3	**4.92 ± 0.3**	6.26 ± 0.3	5.44 ± 0.4
Water	19.21 ± 3.3	5.55 ± 0.9	4.99 ± 0.0	6.69 ± 0.3	4.90 ± 0.3	5.41 ± 0.2	6.95 ± 0.3	**4.16 ± 0.3**
Average	15.64 ± 0.6	7.19 ± 2.4	6.84 ± 0.0	13.80 ± 0.4	5.86 ± 0.3	5.35 ± 0.2	6.35 ± 0.2	**5.13 ± 0.2**



(a) (b) (c)

Fig. 9. For Samson dataset: MCDB-Net estimated endmembers (blue curves) and GT endmembers (red curves). (a) Soil. (b) Tree. (c) Water.

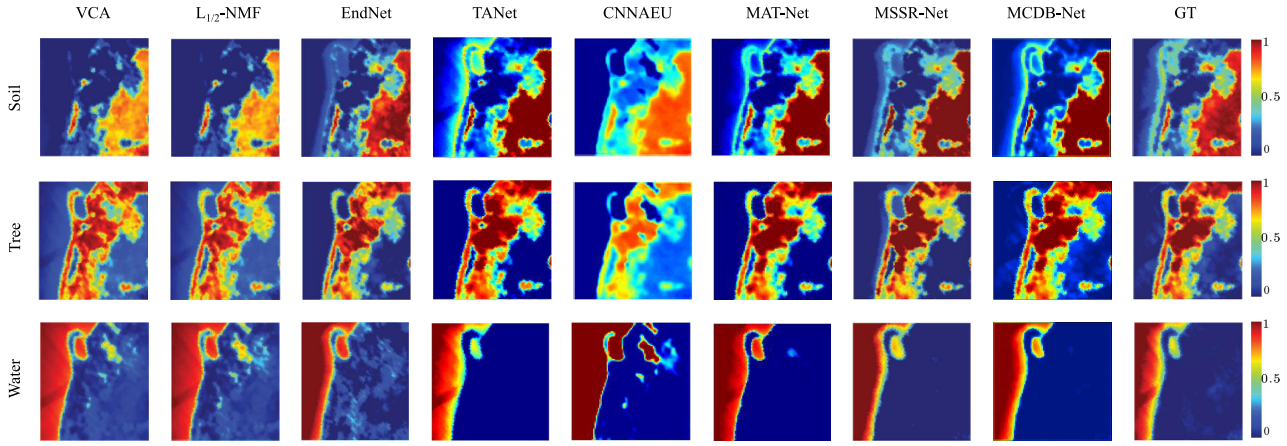


Fig. 10. For Samson dataset: abundance maps estimated by different methods and the relevant GTs.

is consistently excellent, reaching the best level among all the compared methods, and the effect is nearly one percentage point higher than that of other methods. This indicates that MCDB-Net has a more robust and superior performance in the endmember extraction task.

Table VII presents the results of abundance fraction estimation for each unmixing model. It is worth noting that although the CNNAEU model is based on a deep-learning method and is expected to demonstrate superior performance in the HU task compared to traditional methods, the actual results are not as significant as expected. Instead, it performs worse than the $L_{1/2}$-NMF unmixing model. The application of deep-learning methods in the field of HU does not always guarantee the best results. Their performance depends on the comprehensive optimization of the model architecture, training strategy, and data characteristics. It is observed that MCDB-Net achieves the best results in overall abundance estimation and in the abundance estimation of Trees and Water endmembers. By constructing multiview spectral blocks and establishing relationships among these views, this model has realized an innovative approach to collaborative abundance learning from different views. This approach is effective not only because it can deeply explore spectral information from various views but also because it can promote the continuity and consistency of abundance estimation, thus significantly improving the unmixing effect. The successful application of MCDB-Net further emphasizes the importance of adopting multiview and multistrategy comprehensive analysis and processing of hyperspectral data in the field of HU, as well as the necessity of considering the internal structure and diversity of spectral data in model design.

To deeply evaluate the performance of the MCDB-Net in HU, Fig. 9 provides the qualitative analysis results of endmember extraction. It can be seen that there is a high degree of consistency between the endmember spectral curves extracted by MCDB-Net and the actual ground object

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TABLE VIII

FOR THE CUPRITE DATASET, THE EVALUATION METRIC FOR ENDMEMBERS IS SAD ($\times 10^{-2}$). THE OPTIMAL RESULTS OBTAINED FROM THESE EVALUATIONS ARE MARKED IN BOLD

Endmember (SAD)	VCA	$L_{1/2}$-NMF	EndNet	CNNAEU	TANet	MAT-Net	MSSR-Net	MCDB-Net
Alumite	9.62 ± 1.5	14.64 ± 1.8	14.56 ± 2.5	12.34 ± 5.0	12.98 ± 2.5	10.76 ± 0.2	$\mathbf{9.44 \pm 0.7}$	11.09 ± 0.6
Andradite	10.56 ± 2.0	7.86 ± 1.4	8.78 ± 0.5	10.89 ± 5.0	8.12 ± 0.3	9.01 ± 0.8	7.67 ± 0.4	$\mathbf{7.21 \pm 0.7}$
Buddingtonite	11.71 ± 1.5	11.84 ± 5.6	12.45 ± 1.2	12.45 ± 4.0	$\mathbf{10.89 \pm 1.0}$	11.23 ± 1.0	12.77 ± 0.9	12.35 ± 0.8
Dumortierite	9.05 ± 1.5	11.53 ± 3.2	$\mathbf{7.89 \pm 0.5}$	13.12 ± 3.0	9.12 ± 0.8	9.82 ± 1.5	12.21 ± 0.3	8.55 ± 0.9
Kaolinite1	9.84 ± 3.0	9.68 ± 2.3	11.01 ± 1.3	11.23 ± 5.0	11.89 ± 1.0	$\mathbf{9.52 \pm 0.9}$	11.65 ± 0.4	10.99 ± 1.0
Kaolinite2	7.55 ± 1.6	7.24 ± 0.9	13.45 ± 0.5	8.55 ± 3.5	6.12 ± 0.3	6.17 ± 0.3	5.46 ± 0.8	$\mathbf{5.12 \pm 0.9}$
Muscovite	13.66 ± 3.6	25.12 ± 5.5	$\mathbf{9.12 \pm 1.5}$	9.99 ± 3.5	14.78 ± 1.0	11.82 ± 0.4	10.52 ± 0.9	11.68 ± 0.7
Montmorillonite	7.01 ± 1.0	7.46 ± 1.5	8.67 ± 0.3	6.12 ± 3.6	6.02 ± 0.3	6.07 ± 0.9	5.66 ± 0.5	$\mathbf{5.24 \pm 0.5}$
Nontronite	9.21 ± 2.5	9.54 ± 2.3	11.23 ± 0.2	10.56 ± 3.0	8.78 ± 0.5	8.81 ± 0.7	7.55 ± 0.2	$\mathbf{7.33 \pm 0.8}$
Pyrope	11.87 ± 5.0	8.94 ± 5.2	7.45 ± 0.5	8.98 ± 4.0	$\mathbf{6.23 \pm 0.3}$	7.53 ± 1.0	8.23 ± 0.8	7.66 ± 0.4
Sphene	19.30 ± 7.0	7.91 ± 2.5	$\mathbf{7.01 \pm 1.5}$	7.12 ± 3.0	9.78 ± 1.5	8.27 ± 0.3	7.35 ± 1.0	7.56 ± 0.6
Chalcedony	12.56 ± 1.0	13.76 ± 6.3	12.12 ± 0.8	15.16 ± 4.0	$\mathbf{9.34 \pm 1.5}$	13.52 ± 1.1	14.11 ± 0.6	15.12 ± 0.5
Average	10.99 ± 1.5	11.29 ± 1.0	10.31 ± 0.8	10.54 ± 3.0	9.50 ± 1.0	9.38 ± 0.7	9.39 ± 0.6	$\mathbf{9.16 \pm 0.7}$

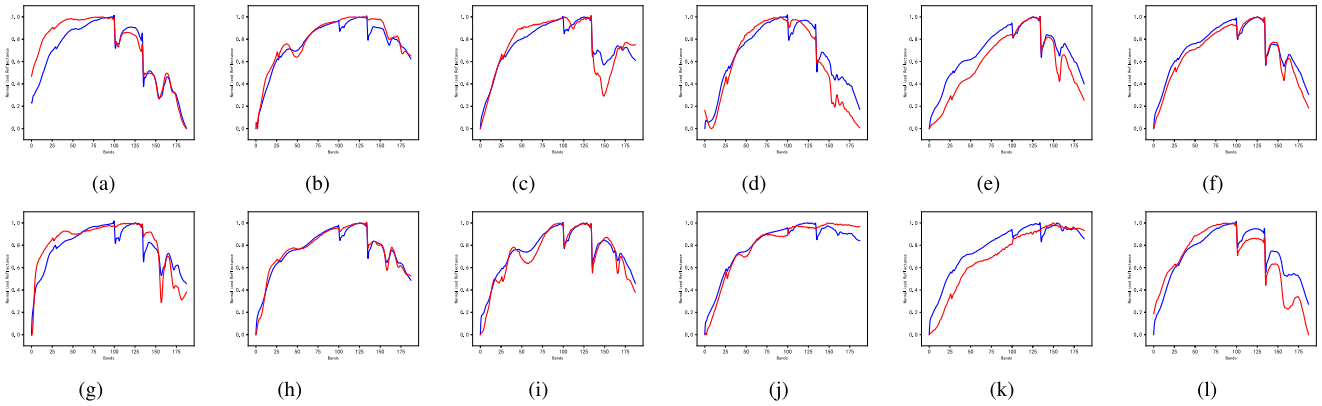


Fig. 11. For Cuprite dataset: MCDB-Net estimated endmembers (blue curves) and GT endmembers (red curves). (a) Alunite. (b) Andradite. (c) Buddingtonite. (d) Dumortierite. (e) Kaolinite1. (f) Kaolinite2. (g) Muscovite. (h) Montmorillonite. (i) Nontronite. (j) Pyrope. (k) Sphene. (l) Chalcedony.

spectral curves, demonstrating the model's remarkable ability to capture the inherent spectral characteristics of ground objects. At the same time, Fig. 10 shows the effect of abundance estimation. By comparing with the true abundance map, it can be observed that the abundance estimation scores of MCDB-Net for each ground object are basically consistent with the true distribution of ground objects. This demonstrates the versatility of the MCDB-Net and its strong adaptability to complex spectral mixing situations, enabling it to accurately reflect the distribution of different ground objects.

4) Experiments on Cuprite Dataset: The endmember extraction results for the Cuprite dataset are presented in Table VIII. As a complex HSI containing twelve endmembers, this dataset poses significant challenges to unmixing algorithms. From the data in the table, it can be observed that the MCDB-Net, designed with a multiview architecture, constructs multiview spectral blocks to comprehensively exploit spectral features from both local and global dimensions. By leveraging this approach, MCDB-Net accurately captures endmember characteristics and achieves optimal performance in endmember extraction. Notably, MCDB-Net demonstrates exceptional performance on individual endmembers such as Andradite, Kaolinite1, Montmorillonite, and Nontronite. These results strongly validate the effectiveness of the proposed multiview spectral block construction and the integration of local–global spectral feature fusion strategies. They highlight the superiority of the multiview architecture-based network in handling endmember extraction tasks for complex hyperspectral data.

Fig. 11 visualizes the endmember extraction performance. It can be clearly observed from the figure that the endmember spectral curves estimated by MCDB-Net show a high degree of coincidence with the real spectral curves. In terms of abundance estimation, due to the absence of real abundances, only the abundance extraction results of MCDB-Net are presented (as shown in Fig. 12). The abundance distribution areas of various materials in the figure are clearly demarcated without overlapping or mixing, intuitively indicating that the model achieves high-quality unmixing results in the current scenario and verifying its effectiveness in the abundance estimation task.

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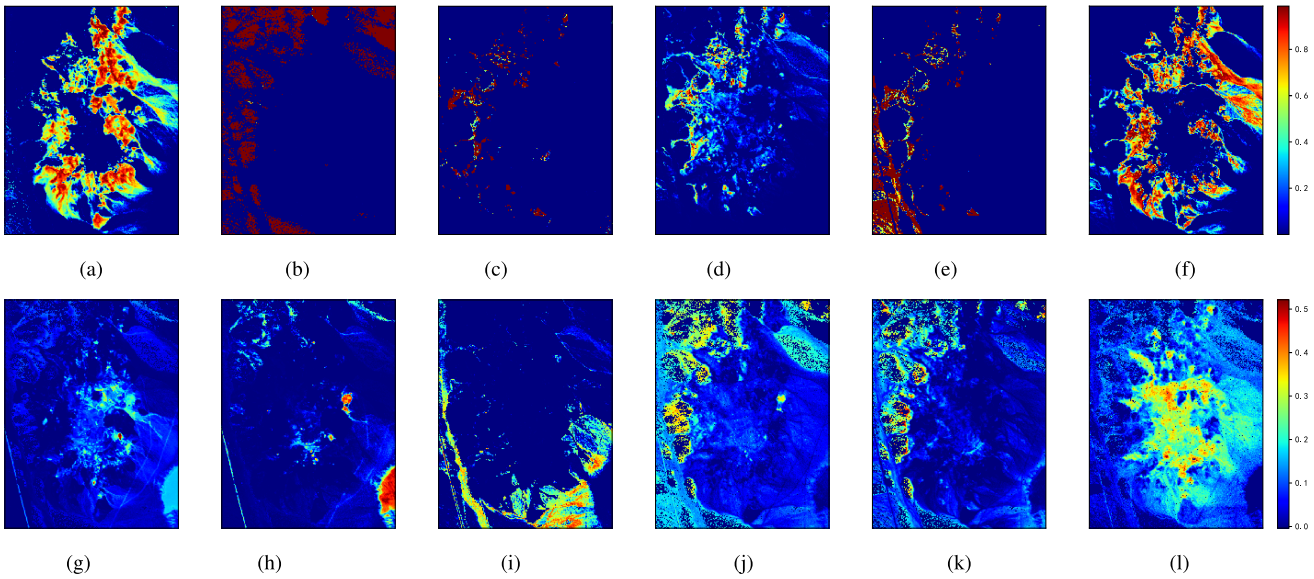


Fig. 12. Abundance maps estimated by MCDB-Net on the Cuprite dataset. (a) Alunite. (b) Andradite. (c) Buddingtonite. (d) Dumortierite. (e) Kaolinite1. (f) Kaolinite2. (g) Muscovite. (h) Montmorillonite. (i) Nontronite. (j) Pyrope. (k) Sphene. (l) Chalcedony.

IV. Conclusion

In this article, a novel MCDB-Net is proposed, which is mainly designed to fully exploit the spectral band features from different perspectives. In this network, the global and local multiview dual-branch structures are trained in parallel, where the global perspective module enhances the features of the full spectral bands using global perspective attention. By constructing a brand-new multiview spectral block and designing a multiview attention model to capture the relationships between local perspectives, the model can fully explore the detailed parts of local spectral features. Furthermore, multiview abundance collaborative learning complements the advantages between different views and eliminates the discontinuity of abundances. The model is experimentally validated on different datasets, strongly demonstrating the superiority of the proposed model in unmixing tasks.

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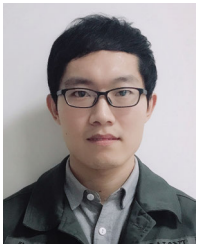
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